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Normal Distribution

Continuous Distribution

Discrete
1) Binomial
2) Poisson
3) Geometric
4) Negative

Continuous
1) Uniform
2) Normal distribution

Central limit theorem

You need to find out the % of value random

Simplify → to use table instead

2% chance

Sample

From a population where we take out a sample the distribution looks generally the same in terms of normal distribution

Mean will be same

From when the distribution is not normal

central limit theorem
→ we take out a sample from a population
→ focus on sample a normal distribution even when the population is not normally distributed

Normal distribution (most important)

$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ $\sigma = \frac{\sigma}{\sqrt{n}}$

* $68-95-99.7$ (Important rule)

Z-score table

$\mu \pm \sigma$ 1 value

$\mu \pm 2\sigma$ 2 value

$\mu \pm 3\sigma$ 3 value

$\mu \pm 1.96\sigma$ 95% confidence interval → 95% of the population lies within this range

$\mu \pm 2.58\sigma$ 99% confidence interval → 99% of the population lies within this range

$\mu \pm 3.29\sigma$ 99.9% confidence interval → 99.9% of the population lies within this range

Q: Mean of students in an exam was normally distributed with value of 74 with $\sigma = 5$ and $\sigma = 10$

1) What is the probability that a student scores less than 84? 15.8
2) Probability that a student scores less than 64? 2.3
3) Probability that the student scores between 64 and 84? 81.9

$\mu = 74$

$\sigma = 10$

$74 + 10 = 84$

$\mu \pm 1\sigma$

$74 - 2 \times 10 = 64$

$\mu - 2\sigma$

$\mu \pm 2\sigma$

$\frac{15.8}{2} = 7.9$

$\frac{2.3}{2} = 1.15$

$\mu \pm 1\sigma$ $\mu \pm 2\sigma$ $\mu \pm 3\sigma$

$Z = \frac{\text{position (How many SD's away from the mean)}}{\sigma}$

$\frac{\text{SD}}{\sigma}$ in the distribution

$\frac{\text{SD}}{\sigma}$ in the sample

$\mu \pm 2\sigma \pm \frac{2\sigma}{\sqrt{n}}$

$Z_{\bar{x}} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$Z = \frac{x - \mu}{\sigma}$

$\frac{\text{Population}}{\text{Sample}}$ $\frac{\text{Population}}{\text{Sample}}$

$Z_{\bar{x}} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$Z_{\bar{x}} = \frac{64 - 74}{\frac{10}{\sqrt{1}}} = \frac{-10}{10} = -0.5$

Approx 30% prob is ≤ 0.5

Note: 0.7881 $-(1-0.7)$ $= 0.2119$

$= 21.2\%$

$Z_{\bar{x}} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$Z = \frac{x - \mu}{\sigma}$

$Z_{\bar{x}} = \frac{84 - 74}{\frac{10}{\sqrt{1}}}$

$Z = \frac{84 - 74}{10} = 1$

$1 - 0.7881 = 0.2119$

$Z_{\bar{x}} = \frac{64 - 74}{\frac{10}{\sqrt{1}}} = \frac{-10}{10} = -1$

$1 - 0.7881 = 0.2119$

$1 - 0.2119 - 0.2119 = 0.5762$

$= 57.62\%$

We know $\mu_{\bar{x}} = \mu = 74$ & $\sigma_{\bar{x}} = \sigma = 10$

$\mu_{\bar{x}} \pm 1\sigma_{\bar{x}}$ $\mu_{\bar{x}} \pm 2\sigma_{\bar{x}}$ $\mu_{\bar{x}} \pm 3\sigma_{\bar{x}}$

$\mu_{\bar{x}} \pm 1.96\sigma_{\bar{x}}$

$Z = \frac{\mu_{\bar{x}} \pm 1.96\sigma_{\bar{x}} - \mu}{\sigma_{\bar{x}}}$

$Z = \frac{\mu_{\bar{x}} \pm 1.96\sigma_{\bar{x}} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$Z = \frac{\mu_{\bar{x}} \pm 1.96\sigma_{\bar{x}} - \mu}{\frac{\sigma}{\sqrt{n}}}$

Z table

$Z = \frac{\mu_{\bar{x}} \pm 1.96\sigma_{\bar{x}} - \mu}{\sigma_{\bar{x}}}$

$\mu_{\bar{x}} \pm 1.96\sigma_{\bar{x}} \rightarrow$ through Z table

$Z = \frac{\mu_{\bar{x}} \pm 1.96\sigma_{\bar{x}} - \mu}{\frac{\sigma}{\sqrt{n}}}$

Q: Daily value of a stock was normally distributed (around) with mean of 200 with σ 30 & σ of 40

1) $P(\text{On a day, stock value} > 230 \text{ value})$ $Z = \frac{230 - 200}{30} = \frac{30}{30} = 1$ $1 - 0.8413 = 0.1587$
2) $P(\text{Stock lies in } 170 \text{ to } 230)$ $0.8413 - 0.0228 = 0.8185$
3) $P(\text{Stock} > 210 \text{ value})$ $Z = \frac{210 - 200}{30} = \frac{10}{30} = 0.33$ We are 99.7% confident that stock value crossed the 210 than

$Z_{210} = \frac{210 - 200}{\frac{30}{\sqrt{1}}} = \frac{10 \times \sqrt{1}}{30} = 0.33$

$\frac{\sigma}{\sqrt{n}}$ $\frac{\sigma}{\sqrt{n}}$ $\frac{\sigma}{\sqrt{n}}$ → $\boxed{0.37}$

$Z_{210} = \frac{210 - 200}{\frac{30}{\sqrt{1}}} = \frac{10}{30} = 0.33$

$Z_{210} = \frac{210 - 200}{\frac{30}{\sqrt{1}}} = \boxed{1} = \boxed{0.8413}$

$1 - 0.8413 = 0.1587$

P-value (Very imp)

Q: We are testing whether a new medicine reduces blood pressure or not

Null hypothesis

H_0 (Null hypothesis) The medicine has no effect (Probability)

H_1 (Alternate) The medicine works (reduction of)

We have the medicine to 30 patients

Avg reduction in blood pressure $= \frac{\text{Sum of all}}{\text{Total}}$

$30 = \frac{\text{Sum of all}}{\text{Total}}$

Compute Z-Test

$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$\bar{x}$ = Sample mean
μ = Population mean
σ = Sample standard
n = 30 of samples

$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$P(Z \leq 0.8) = 0.8813$
$P(Z \geq 0.8) = 0.8813$

P-value

[Probability that this change of blood pressure was by chance = $\frac{\text{P-value}}{2}$]

What is P-value

→ The probability of value to be as extreme as $\bar{x}$ is a chance

Low P-value

P-value

What are the chances they be

Why chance that they were picked by we to the 30 that lowest medicine

If P-value is close to zero

$P(Z \geq 3) = \frac{P(Z \geq 3)}{2}$

$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

We are very confident that the medicine lowered

Q: We are testing whether a sample company's medicine works or not

H_0 (Null hypothesis) Medicine has no effect

H_1 (Alternate) Medicine works

Avg medicine $= \frac{\text{Sum}}{\text{Total}}$

$30 = \frac{\text{Sum}}{\text{Total}}$

Did medicine work

$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

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$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$P(Z \geq 3) = \frac{P(Z \geq 3)}{2}$

P-value

If P is low, accept the alternate hypothesis

If P is high, accept null hypothesis

NH-effect

Q: We suspect that a coin is biased

H_0 : Coin is fair

H_1 : Coin is not fair (biased)

We test this coin 10 times we get 8 heads

$P(X \geq 8) = P(8) + P(9) + P(10)$

$= {}^{10}C_8 p^8 q^2 + {}^{10}C_9 p^9 q^1 + {}^{10}C_{10} p^{10} q^0$

combinatorial factor

$= 10 \times \frac{1}{2}^8 \times \frac{1}{2}^2 + 10 \times \frac{1}{2}^9 \times \frac{1}{2}^1 + \frac{1}{2}^{10}$

$= 45 \times \frac{1}{2}^{10}$

$= 0.0439$

$= 0.0439$

P-value $= 4.39\%$

that we can confidently say that coin is biased (H_1)

If $P < 0.05$ then accept H_1 (Alternate hypothesis)

If $P \geq 0.05$ then accept H_0 (Null hypothesis)

Coin was biased in 8 times

$P = ?$

$P(X \geq 8) = \frac{P(X \geq 8)}{2} = \frac{P(8) + P(9) + P(10)}{2}$

$= \frac{{}^{10}C_8 p^8 q^2 + {}^{10}C_9 p^9 q^1 + {}^{10}C_{10} p^{10} q^0}{2}$

combinatorial factor

$= \frac{10 \times \frac{1}{2}^8 \times \frac{1}{2}^2 + 10 \times \frac{1}{2}^9 \times \frac{1}{2}^1 + \frac{1}{2}^{10}}{2}$

$= \frac{45 \times \frac{1}{2}^{10}}{2}$

$= \frac{10 \times 0.0439}{2}$

P-value $= 2.19\%$

$P \leq 0.05$ (We can accept our hypothesis)

If medicine has effect

If the effect is not where we will say that we reject for medicine

$P(X \geq 8 \text{ or } X \leq 2 \text{ high-impact heads}) \leq 0.05$

P-value → Probability to have

$P \leq 0.05$ → we can reject our hypothesis

$P \geq 0.05$ → we can't reject our hypothesis

Alternate $\rightarrow 95\%$

P value